

Structural LQI–PIDF equivalence and constrained weight tuning for a wide-range Buck converter

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Abstract

Metaheuristic tuning of PID-type controllers for power converters usually searches gain spaces that carry no stability guarantee, and its results are rarely checked against switching effects. This paper shows that, for a non-ideal Buck converter, the linear–quadratic–integral (LQI) law has an exact realization as a two-degree-of-freedom PIDF that measures only the output voltage. When its derivative pole is placed on the capacitor ESR zero, the filter state reconstructs the capacitor voltage along every trajectory. The PIDF and the full-state LQI therefore coincide under saturation, anti-windup and discontinuous conduction. They differ only through an explicit output-current term, output-network mismatch and sampling, and a first-order-hold discretization reduces the sampling gap by a factor of about 40. The controller is tuned by searching the LQI weights, not the gains, so every candidate is LQ-optimal and carries guaranteed margins and a certified pole region. A constrained, multi-fidelity search includes short switched-model simulations to reject coexisting switching limit cycles that averaged models miss. On a 48 V converter commanded from 1 to 46 V, six metaheuristics converge to the same optimum. An unconstrained gain search reaches the same controller, but 42 % of its feasible solutions are not LQ-optimal. On a switched replica of the reference Simulink model, the tuned controller eliminates overshoot and respects the device current rating, whereas standard PI, PID, PIDF, LQR, LQG and LQI designs exceed the specification by factors of 6 to 30.

Keywords: Buck converter, LQI control, 2-DOF PIDF, structural equivalence, metaheuristic optimization, multi-fidelity constraints, switching limit cycles

1. Introduction

Digitally controlled DC–DC converters are routinely tuned by metaheuristics: particle swarms, genetic algorithms, bee colonies and grey-wolf variants search the gains of a PI, PID or filtered PID (PIDF) against a tracking index [18, 26, 28]. Two weaknesses recur in that literature, and both are visible to an industrial reader. First, the searched object is a vector of gains with no structural meaning, so a good gain vector carries no stability or robustness certificate and the search space is sized by guesswork. Second, the evidence is usually produced on the same averaged model that drives the search, so switching ripple, duty saturation and discontinuous conduction, which decide whether a tuning is admissible on the real power stage, never enter the comparison [23, 15].

This paper addresses both points on one concrete, audited plant: a non-ideal asynchronous Buck converter supplied from a fixed 48 V source and commanded over the whole attainable range 1 V to 46 V. The design is *formula first*. The searched object is not a gain vector but the set of Bryson excursions that define a linear–quadratic–integral (LQI) cost; every candidate is solved from the algebraic Riccati equation and then realized *exactly* as a two-degree-of-freedom PIDF (2-DOF PIDF) that reads the output voltage only. Every evaluated point is therefore a genuine LQI law, with the classical LQR return-difference guarantees, and at the same time a controller that a practitioner can implement in a standard PID block with setpoint weights.

The central theoretical observation is stronger than a transfer-function identity. We show that the derivative filter of the realizing PIDF, when its pole is placed at the equivalent-series-resistance (ESR) zero $N_C = 1/(r_C C)$ and its state is initialized consistently, reconstructs the capacitor voltage *exactly*, whatever happens in the inductor branch. Consequently the output-feedback PIDF and the full-state LQI generate identical trajectories under duty saturation, anti-windup activity, discontinuous conduction and any mismatch of L , r_L , R_{on} , r_d or V_f ; only an unmeasured load current, a mismatch of the output network (R, r_C, C) , or sampling break the equivalence. This turns a nominal, small-signal equivalence into a structural statement that can be checked on a simulator to machine precision, and it tells the practitioner exactly which sensors the PIDF saves and at what price.

The optimization layer compares six equal-budget global operators—PSO, GA, ABC and three project variants pAEABC, pIGWO and pIGWO–DLH—each followed by a common Gaussian-process refinement stage (re-

fineBO), over 30 matched seeds. The seven algorithms are not the contribution by themselves; the question they answer is whether the *LQI-weight parameterization* helps them. We therefore run the identical protocol on five other controller families that share the evaluator, the budget and the specifications: the full-state twin of the design (same weights, measured i_L), the same 2-DOF PIDF searched directly in gain space, a PIDF with free filter pole and setpoint weights, a PID and a PI. Six fixed-rule comparators—PI, PID, PIDF (Type-III rule), LQR, LQG and LQI—complete the picture, and every frozen controller is replayed on a switched replica of the reference Simulink/Simscape model on the complete 24→1→46 V mission, with a closed energy balance.

A previous version of this protocol accepted candidates on an averaged model guarded by an analytical switching-ripple bound, and four of six branch winners failed on the switched model. We reproduce that failure mode, show that it is caused by coexisting switching limit cycles whose selection depends on the preceding transient, and replace the analytical bound by three *measured* inequalities obtained from a 34 ms switched transition sequence inside each evaluation, a multi-fidelity constraint that roughly doubles the cost of a candidate. Feasibility is also stated in absolute engineering units (overshoot, load deviation, device current, steady error) instead of ratios to a reference controller, so that every family is judged against the same specification.

The contributions are:

1. a formula-first chain from the non-ideal converter equations to the exact output-feedback realization of any LQI law as a 2-DOF PIDF, with all analytical quantities (equilibria, transfer functions, Riccati gain, LQR margins, a common- P D-region certificate over the operating polytope) instantiated and independently recomputed;
2. a structural equivalence theorem (Proposition 2) showing that the PIDF filter state equals the capacitor voltage along every trajectory, with its exact domain of validity, a closed-form expression of the gap to the full-state law, and the discretization that preserves it under sampling;
3. a constrained co-design problem in the space of LQI weights with 33 inequalities in engineering units, three of which are measured on the switched model during the search, and the evidence (two reported pilots) of why an analytical ripple bound or a steady switched screen is not enough;

- 75 4. a matched, equal-budget comparison of seven algorithms across six con-
76 troller families, showing that the LQI coordinates do not speed up the
77 search but confine it, at no cost in optimality, to the LQ-optimal man-
78 ifold on which the gain-space optimum also lies;
- 79 5. a switched-model validation on the reference mission against PI, PID,
80 PIDF, LQR, LQG and LQI, reported specification by specification.

81 Section 3 derives the model and describes the switched replica. Section 4
82 states the LQI synthesis and the equivalence results. Section 5 defines the
83 co-design problem and the algorithms. Section 6 reports the results and
84 Section 7 concludes. All code, including a MATLAB S-function that runs
85 the same control layer in the reference Simulink model, is released with the
86 paper.

87 2. Related work and positioning

88 *Model-based converter control.* State-space averaging supplies the standard
89 low-cost converter model [23, 15]; non-ideal averaged plants with inductor
90 and capacitor ESR, switch resistance and diode drop have been used to tune
91 implementable low-order controllers with hardware validation [31], and ro-
92 bust nonlinear or observer-based laws extend disturbance rejection [1, 32, 7].
93 Convex methods—MPC solved as a quadratic program [22], LMI regional
94 pole placement [9]—give certificates relative to their model class but do not
95 make fixed-structure PIDF tuning convex once saturation, discontinuous con-
96 duction and event-wise requirements are included.

97 *LQ design and PID structure.* The LQR return-difference inequality guaran-
98 tees an infinite upward gain margin, a gain reduction margin of one half and
99 60° of phase margin for the full-state loop [19, 29, 3]; the guarantee is not
100 inherited by observer-based implementations [13], and loop-transfer recovery
101 trades it against observer bandwidth [14]. Bryson’s rule converts admissible
102 excursions into diagonal weights [8]. Setpoint weighting and back-calculation
103 anti-windup are the standard means of shaping the command response of a
104 PID and of handling actuator saturation [5, 4, 6]. The connection exploited
105 here—that the full-state LQI law of a second-order plant with an output-
106 network zero is a filtered PID of the measured output—is known in nominal
107 transfer-function form; what is new is its trajectory-level validity and the use
108 of the Riccati weights, not the gains, as the search space.

Table 1: Positioning with respect to representative Buck control and tuning studies. “Not reported” refers to the stated scope of each work only.

Study	Plant evidence	Controller / search	Relative to the present question
Siddhartha et al. [31]	non-ideal Buck, dSPACE	IMC-PID, direct	no repeated solver comparison, no switched energy audit
Izci et al. [18]	averaged Buck	PID, AEO + Nelder-Mead	gain-space search, no structural certificate, no switched admissibility
Nanyan et al. [26]	averaged Buck	PID, improved SCA	gain-space search, single fidelity
Siano and Citro [30]	Buck prototype	fuzzy, MO-PSO	no equal-budget solver comparison
Abdelmalek et al. [1]	Buck, experimental	observer-based nonlinear, PSO	tracking/robustness focus, not a matched co-design
Coutinho et al. [10]	multiloop process	PID, Bayesian optimization	not a power converter
Present work	audited non-ideal Buck, 1–46 V, switched replica with energy balance	LQI weights $\rightarrow$ exact 2-DOF PIDF; 6 global operators + refineBO; 6 families	structural equivalence theorem; 33 inequalities (3 measured on the switched model); switched validation vs. PI, PID, PIDF, LQR, LQG, LQI

109 *Metaheuristic tuning.* PSO [21], genetic algorithms, ABC [20] and GWO
110 [24] with improved variants [16, 25, 2] are widely used to tune converter con-
111 trollers [18, 26, 35]; surrogate-based Bayesian optimization is attractive when
112 evaluations are expensive [33, 10]. Constraint handling by Deb’s feasibility
113 rules avoids penalty weights [11], and matched-seed non-parametric statistics
114 are the accepted way to compare stochastic solvers [12]. A weighted scalar
115 index does not produce a Pareto front [28]; we therefore report a constrained
116 scalar problem and do not claim a multi-objective characterization.

117 *Switching effects in digital control.* In digitally controlled PWM converters,
118 the sampled switching ripple and the duty resolution can sustain limit cycles
119 that an averaged model does not predict [27]. This is the mechanism behind
120 the rejection of averaged-feasible candidates reported in Section 6.5, and the
121 reason why three inequalities of the present problem are measured on the
122 switched model.

123 Table 1 positions the study. The gap is not a new metaheuristic but the
124 conjunction of (i) a structurally meaningful search space whose every point
125 carries an LQ certificate and an exact output-feedback realization, (ii) a fea-
126 sibility set stated in engineering units and partly measured on the switched
127 plant, (iii) matched equal-budget comparisons that separate the parameteri-
128 zation from the optimizer, and (iv) a closed-energy-balance switched valida-
129 tion on the full commanded range.

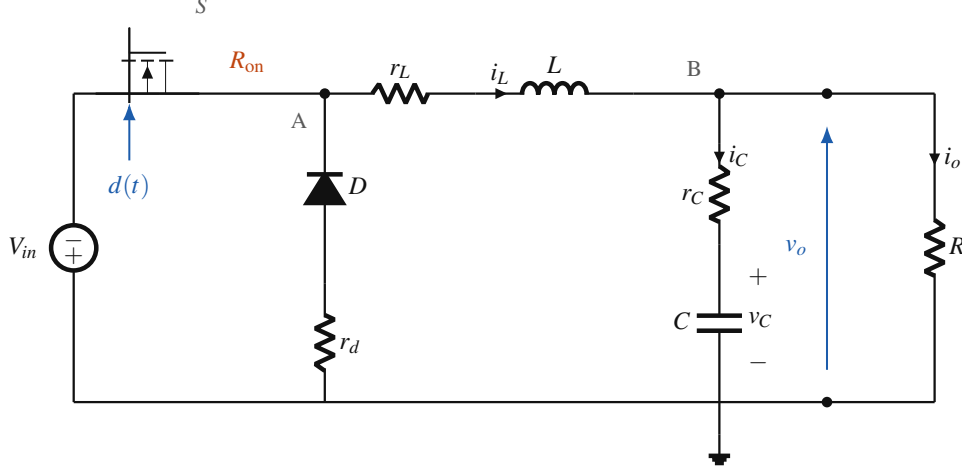


Figure 1: Non-ideal asynchronous Buck converter and adopted conventions.

130 3. Non-ideal Buck converter: from the circuit to the switched 131 replica

132 3.1. Circuit and conventions

133 Fig. 1 shows the asynchronous converter. The state is $x = [i_L \ v_C]^\top$; i_L
134 flows from the switching node to the output, i_C flows downward in the capac-
135 itor branch and the load disturbance i_p is positive when it draws additional
136 current. The source is fixed, $V_{in} = 48$ V. The high-side switch is an IRF540N
137 represented by its static on-resistance R_{on} ; the freewheel diode follows the
138 affine conduction law of the STPST10H100SF data sheet, $v_D = V_f + r_d i_D$
139 [17, 34]. Table 2 lists the values, all decoded from the workspace of the
140 reference Simulink model.

141 3.2. Averaged model and loss-aware equilibrium

142 The output node and the ESR give

$$i_L = i_C + \frac{v_o}{R} + i_p, \quad v_o = v_C + r_C i_C, \quad (1)$$

143 hence

$$i_C = \frac{R(i_L - i_p) - v_C}{R + r_C}, \quad v_o = \frac{R}{R + r_C} v_C + \frac{R r_C}{R + r_C} (i_L - i_p). \quad (2)$$

Table 2: Electrical and numerical parameters (reference model workspace).

Quantity	Value	Role
V_{in}	48 V, constant	source
L, r_L	100 μ H, 0.10 Ω	inductor branch
C, r_C	100 μ F, 0.05 Ω	capacitor branch
R	10 Ω	nominal load
R_{on}	0.05 Ω	IRF540N static surrogate
V_f, r_d	0.42 V, 0.020 Ω	STPST10H100SF affine law
R_s ($C_s = \infty$)	100 k Ω	resistive snubber across the switch
$R_{d,body}$	0.01 Ω	MOSFET body diode
f_s	50 kHz	PWM carrier
T_s	10 μ s	controller sampling ($= 1/(2f_s)$)
$[d_{\min}, d_{\max}]$	[0.02, 0.98]	total duty saturation

144 With the switch on, $L\dot{i}_L = V_{in} - (r_L + R_{on})i_L - v_o$; with the diode conducting,
 145 $L\dot{i}_L = -V_f - (r_L + r_d)i_L - v_o$; in both cases $C\dot{v}_C = i_C$. Duty averaging yields

$$L\dot{i}_L = dV_{in} - [r_L + dR_{on} + (1-d)r_d]i_L - (1-d)V_f - v_o. \quad (3)$$

146 At equilibrium $I_L = V_o/R + I_p$ and the loss-aware duty is

$$d_{ff} = \frac{V_o + (r_L + r_d)I_L + V_f}{V_{in} - (R_{on} - r_d)I_L + V_f}. \quad (4)$$

147 Every controller of this paper uses (4) with nominal parameters and $I_p = 0$
 148 as feedforward; the feedforward never sees the simulated plant.

149 3.3. Small-signal model and transfer functions

150 Let $\gamma = R/(R + r_C)$, $\rho_0 = r_L + r_d + d_0(R_{on} - r_d)$ and $E_d = V_{in} +$
 151 $V_f - (R_{on} - r_d)I_{L,0}$. Linearizing (2)–(3) gives $\delta\dot{x} = A\delta x + B_d\delta d + B_p\delta i_p$,
 152 $\delta v_o = C_y\delta x + D_p\delta i_p$ with

$$A = \begin{bmatrix} -(\rho_0 + \gamma r_C)/L & -\gamma/L \\ \gamma/C & -\gamma/(RC) \end{bmatrix}, \quad B_d = \begin{bmatrix} E_d/L \\ 0 \end{bmatrix}, \quad C_y = [\gamma r_C \quad \gamma], \quad (5)$$

153 $B_p = [\gamma r_C/L \quad -\gamma/C]^\top$ and $D_p = -\gamma r_C$. The control-to-output transfer
 154 function is

$$G_{vd}(s) = \frac{E_d R(1 + r_C C s)}{\Delta(s)}, \quad (6)$$

$$\Delta(s) = LC(R + r_C)s^2 + [L + C((R + r_C)\rho_0 + R r_C)]s + R + \rho_0,$$

Table 3: Equilibria, monic $\Delta(s)$, G_{vd} numerator and stationary conduction screen at the three commanded levels (nominal load, $i_p = 0$). $\omega_z = 1/(r_C C) = 2 \times 10^5$ rad/s at every level.

V_o (V)	$I_{L,0}$ (A)	d_0	ρ_0 (Ω)	E_d (V)	$\Delta(s)/(LC(R + r_C))$	G_{vd} numerator	$\Delta I_{L,pp}$ (A)	I_{crit} (A)	$I_L - I_{crit}$ (A)	R_{crit}/R
1	0.1	0.029576	0.120887	48.417	$s^2 + 2701.41s + 1.00705 \times 10^8$	$2.40881 \times 10^4 s + 4.81761 \times 10^9$	0.1941	0.0970	0.0030	1.030
24	2.4	0.511045	0.135331	48.348	$s^2 + 2845.85s + 1.00849 \times 10^8$	$2.40537 \times 10^4 s + 4.81075 \times 10^9$	2.3470	1.1735	1.2265	2.045
46	4.6	0.972868	0.149186	48.282	$s^2 + 2984.40s + 1.00987 \times 10^8$	$2.40209 \times 10^4 s + 4.80418 \times 10^9$	0.2496	0.1248	4.4752	36.856

155 with the ESR zero $\omega_z = N_C = 1/(r_C C) = 2 \times 10^5$ rad/s. Table 3 instantiates
 156 (4)–(6) at the three commanded levels; the closed form and a state-space
 157 conversion agree to machine precision. At the 24 V synthesis point, (A, B_d)
 158 and (A, C_y) have rank two and the integral-augmented pair has rank three.

159 3.4. Conduction mode across the range

160 For a duty d_{ff} the inductor ripple, the boundary current and the bound-
 161 ary load are

$$\Delta I_{L,pp} = \frac{V_o(1 - d_{ff})}{Lf_s}, \quad I_{crit} = \frac{\Delta I_{L,pp}}{2}, \quad R_{crit} = \frac{2Lf_s}{1 - d_{ff}}. \quad (7)$$

162 The stationary screen (Table 3) is positive at every level but not uniform:
 163 at 1 V the current margin is 2.96 mA (3.05 % of I_{crit}) and R_{crit} is only 1.030
 164 times the nominal load. The switched model is more pessimistic than this
 165 screen: over the steady 1 V window it spends 15.00 % of the time in discon-
 166 tinuous conduction at 60.57 % efficiency, because the diode drop is 42 % of
 167 the commanded output. Discontinuous conduction is thus a nominal regime
 168 of the mission, and every controller is validated in it.

169 3.5. Switched replica of the reference model

170 The acceptance model (Fig. 2) is the Simulink/Simscape Electrical model
 171 `buck_fo_lqi_fixed_reference_v6_1_46V_R2022a.slx`. For the campaign we
 172 use a replica written from its block parameters (`cmo/sim.py`): a resistive
 173 switch with snubber R_s and a body diode $R_{d,body}$; the affine freewheel diode,
 174 which blocks reverse current so that discontinuous conduction arises by it-
 175 self (the 1 ns snubber-leakage branch is solved quasi-statically); a continuous
 176 sawtooth carrier whose edges are located exactly inside the integration step;
 177 a controller sampled at T_s with a one-sample output delay, exactly as the
 178 S-function of the reference model. The step is 0.2 μ s (100 steps per switching
 179 period) with fourth-order Runge–Kutta integration.

180 The energy boundary is the one declared in the model workspace:

$$\begin{aligned} P_{in} &= V_{in}i_{in}, & P_u &= \frac{v_o^2}{R} + v_o i_p, \\ P_{loss} &= r_L i_L^2 + r_C i_C^2 + v_{SW} i_{SW} + v_D i_D, \end{aligned} \quad (8)$$

181 the snubber dissipation being contained in $v_{SW}i_{SW}$. The three energies are
 182 integrated with the same Runge–Kutta stages as the states, so the residual
 183 $E_{in} - E_u - E_{loss} - \Delta E_{stored}$ measures integration consistency only. It stays
 184 below $4.5 \times 10^{-6} \%$ of the input energy for every replay reported here.

185 The replica reproduces the controller-independent figures of the reference
 186 model: 15.00 % of discontinuous conduction and 60.57 % efficiency in the 1 V
 187 window (15.27 % and 60.51 % in the Simulink audit of the previous version
 188 of this study). Its discrete LQI comparator gain, synthesized by the rule
 189 of Section 5.5, equals the gain stored in the model workspace to all printed
 190 digits, and the fixed PI and LQI comparators reproduce the Simulink over-
 191 shoot and peak current to within 2 % (Section 6.5). A MATLAB S-function
 192 (`matlab/cmo_sfun.m`) implements the same control layer line by line, and
 193 `matlab/run_cmo_validation.m` replays every frozen controller on the refer-
 194 ence `.slx` without modifying it.

195 4. LQI synthesis in weight space and its exact 2-DOF PIDF real- 196 ization

197 4.1. Integral augmentation and Bryson weights

198 Let $z(t) = \int_0^t (r - v_o) d\tau$. Around the synthesis equilibrium ($V_o = 24$ V)
 199 the augmented plant is

$$\dot{x}_a = A_a x_a + B_a \delta d, \quad A_a = \begin{bmatrix} A & 0 \\ -C_y & 0 \end{bmatrix}, \quad B_a = \begin{bmatrix} B_d \\ 0 \end{bmatrix}, \quad x_a = \begin{bmatrix} \delta x \\ z \end{bmatrix}, \quad (9)$$

200 and the LQI law $\delta d = -K x_a$, $K = [K_{ix} \ K_{vx} \ K_z]$, minimizes $\int_0^\infty (x_a^\top Q x_a +$
 201 $R_d \delta d^2) dt$ with

$$A_a^\top P + P A_a - P B_a R_d^{-1} B_a^\top P + Q = 0, \quad K = R_d^{-1} B_a^\top P. \quad (10)$$

202 Bryson’s rule assigns $Q = \text{diag}(\Delta_i^{-2}, \Delta_v^{-2}, (\Delta_v/\omega_B)^{-2})$ and $R_d = \Delta_d^{-2}$: Δ_i
 203 and Δ_v are admissible current and voltage excursions and the integral state,
 204 in Vs, is scaled by a target bandwidth ω_B . The duty excursion is fixed at

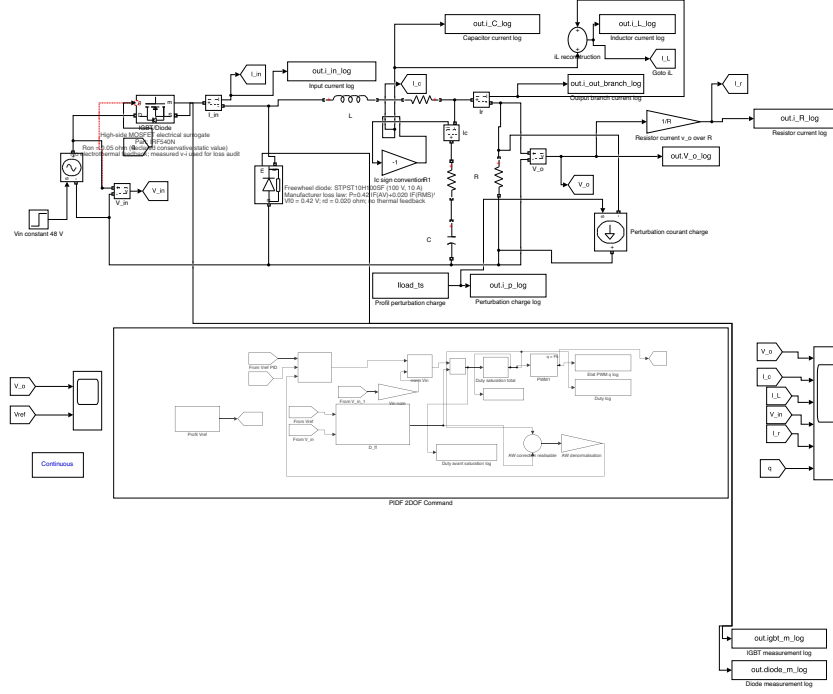


Figure 2: Reference Simulink/Simscape Electrical switching model (fixed 48 V source, IRF540N surrogate, STPST10H100SF diode, resistive snubber, continuous PWM at 50 kHz, sampled controller subsystem and energy loggers). Only the controller S-function is replaced when a candidate is replayed.

its physical value $\Delta_d = (d_{\max} - d_{\min})/2 = 0.48$, which removes the scale
 redundancy $(Q, R_d) \rightarrow (\alpha Q, \alpha R_d)$ of (10). The decision vector of the co-
 design is

$$\xi = [\log_{10} \Delta_i, \log_{10} \Delta_v, \log_{10} \omega_B, \log_{10} K_b]^\top, \quad (11)$$

where K_b is the back-calculation rate of the anti-windup, which lies outside
 the linear theory and is searched alongside the weights. The normalization
 record ξ_0 uses $\Delta_i = 15$ A, $\Delta_v = 46$ V (the largest commanded output) and
 $\omega_B = K_b = 2\pi \cdot 1500$ rad/s.

4.2. Frequency-domain realization

For $i_p = 0$ the output network gives, in the Laplace domain, $v_C = v_o/(1 +$
 $r_C Cs)$ and $i_L = v_o/R + C s v_o/(1 + r_C Cs)$. Substituting into the state part of

215 the LQI law,

$$\frac{K_{ix}i_L + K_{vx}v_C}{v_o} = \frac{K_{ix}}{R} + \frac{K_{ix}Cs + K_{vx}}{1 + r_CCs} = K_p + K_d \frac{N_Cs}{s + N_C}, \quad (12)$$

216 which exposes one constant and one filtered derivative term.

217 **Proposition 1** (Nominal realization). *The output-error compensator gener-*
 218 *ated by the full-state LQI law is the 2-DOF PIDF*

$$d = d_{ff}(r) + K_p(br - v_o) + \frac{K_i}{s}(r - v_o) + K_d \frac{N_Cs}{s + N_C}(cr - v_o), \quad (13)$$

219 with

$$\begin{aligned} K_p &= K_{vx} + \frac{K_{ix}}{R}, & K_i &= -K_z, & K_d &= C(K_{ix} - r_C K_{vx}), \\ N &= N_C = \frac{1}{r_C C}, & b &= 1, & c &= 0. \end{aligned} \quad (14)$$

220 *Proof.* Split the right-hand side of (12) into its zero-frequency value $K_{vx} +$
 221 K_{ix}/R and the remainder $C(K_{ix} - r_C K_{vx})N_Cs/(s + N_C)$. The equilibrium
 222 map $x_{eq}(r) = [r/R \ r]^\top$ contributes $+(K_{ix}/R + K_{vx})r = K_p r$, hence $b = 1$;
 223 the state-feedback law has no derivative of the reference, hence $c = 0$; the
 224 integral state contributes $-K_z z$, hence $K_i = -K_z$. $\square$

225 Because $N_C = 2 \times 10^5 \text{ rad/s} < \pi/T_s = 3.14 \times 10^5 \text{ rad/s}$, the exact filter
 226 pole is implementable at the adopted sampling period, and no projected
 227 surrogate is needed: every candidate of the search is realized with its own
 228 exact pole.

229 4.3. Structural (trajectory-level) equivalence

230 Proposition 1 is a statement about transfer functions, i.e. about the linear
 231 nominal plant with consistent initialization. The next result shows that the
 232 realization is in fact exact along every trajectory of the *nonlinear* converter,
 233 because it only uses the output network, never the inductor branch. Write
 234 the PIDF of (13) in state form,

$$\begin{aligned} \tilde{d} &= d_{ff}(r) + K_p(br - v_o) + x_I + K_d N(cr - v_o - x_F), & d &= \text{sat}(\tilde{d}), \\ \dot{x}_I &= K_i(r - v_o) + K_b(d - \tilde{d}), & \dot{x}_F &= N(cr - v_o - x_F), \end{aligned} \quad (15)$$

235 and the LQI with the same feedforward and anti-windup as

$$\begin{aligned} \tilde{d} &= d_{ff}(r) - K_{ix}\left(i_L - \frac{r}{R}\right) - K_{vx}(v_C - r) - K_z z, \\ \dot{z} &= (r - v_o) + \frac{K_b}{|K_z|}(d - \tilde{d}). \end{aligned} \quad (16)$$

Proposition 2 (Structural equivalence). *Let r be piecewise constant, let the PIDF use (14) with $c = 0$, $N = N_C$ computed from the true (r_C, C) , and let $x_F(0) = -v_C(0)$ and $x_I(0) = -K_z z(0)$. Then, along every trajectory of the converter—averaged or switched, in continuous or discontinuous conduction, with or without active saturation and anti-windup, and for any values of L , r_L , R_{on} , r_d , V_f and V_{in} :*

- (i) $x_F(t) = -v_C(t)$ for all $t \geq 0$;
- (ii) the two unsaturated commands satisfy

$$\tilde{d}_{LQI}(t) - \tilde{d}_{PIDF}(t) = -K_{ix} \left(i_o(t) - \frac{v_o(t)}{R} \right), \quad (17)$$

where $i_o = i_L - i_C$ is the actual output-branch current and R the nominal load;

- (iii) if $i_o \equiv v_o/R$ (no load disturbance, nominal load), the two closed loops generate identical trajectories.

Proof. (i) The capacitor branch gives $v_o = v_C + r_C C \dot{v}_C$ exactly, whatever flows into the output node, so $\dot{v}_C = N_C(v_o - v_C)$. Let $\varepsilon = x_F + v_C$. Between reference steps, with $c = 0$, $\dot{\varepsilon} = N_C(-v_o - x_F) + N_C(v_o - v_C) = -N_C \varepsilon$, and ε is continuous across steps because $c = 0$. Hence $\varepsilon(0) = 0$ implies $\varepsilon \equiv 0$. (ii) With (i), the derivative term of (15) is $K_d N_C(v_C - v_o) = -K_d N_C r_C i_C = -(K_{ix} - r_C K_{vx}) i_C$. In (16) substitute $i_L = i_o + i_C$ and $v_C = v_o - r_C i_C$: $-K_{ix}(i_L - r/R) - K_{vx}(v_C - r) = K_p(r - v_o) - (K_{ix} - r_C K_{vx}) i_C - K_{ix}(i_o - v_o/R)$. With $b = 1$ and $x_I = -K_z z$ the difference of the two commands is (17). (iii) If the right-hand side of (17) vanishes, both laws issue the same $\tilde{d}$ and d at every instant. Then $\frac{d}{dt}(x_I + K_z z) = K_i(r - v_o) + K_b(d - \tilde{d}) - |K_z|(r - v_o) - K_b(d - \tilde{d}) = 0$ because $K_z < 0$, so $x_I = -K_z z$ is preserved, and the two closed-loop systems coincide; uniqueness of solutions gives identical trajectories. $\square$

Three practical consequences follow. First, the PIDF does not approximate the LQI on the nonlinear converter; it *is* the LQI whenever the load current is the nominal one, and the saturation, anti-windup and conduction-mode effects that invalidate small-signal arguments do not break it. Second, the price of removing the current sensor is stated exactly by (17): the full-state law contains an instantaneous proportional feedback $-K_{ix}(i_o - v_o/R)$ on the unmodelled output current, which the PIDF must recover through its integral action. Third, the equivalence is lost by exactly three mechanisms: a load disturbance or load mismatch (through (17)), a mismatch of (r_C, C)

269 (through $N \neq N_C$ in (i)), and sampling, because the discrete filter recon-
 270 structs v_C only between samples while the sampled LQI reads i_L with its
 271 switching ripple. Section 6.2 verifies each case.

272 4.4. Guarantees and their scope

273 *LQR margins.* For the full-state augmented loop $L_K(s) = K(sI - A_a)^{-1}B_a$
 274 the return-difference identity gives $|1 + L_K(j\omega)| \geq 1$, hence the gain margin
 275 $(1/2, \infty)$ and a phase margin of at least 60° [29, 3]. By Proposition 2(iii)
 276 these margins are those of the PIDF loop at the input of the nominal linear
 277 plant; they do not survive sampling or an observer [13].

278 *Operating-polytope certificate.* The augmented closed-loop matrix is affine
 279 in $(\rho_0(d_0), E_d(I_{L,0}))$. Over $v_o \in [1, 46]$ V and $i_p \in [0, 0.5]$ A the rectangle
 280 $d_0 \in [0.0295764, 0.9744132]$, $I_{L,0} \in [0.1, 5.1]$ A has four vertices $A_{cl,k}$. A
 281 common $P \succ 0$ with

$$A_{cl,k}P + PA_{cl,k}^\top + 2\alpha P \prec 0, \quad \begin{bmatrix} \sin \theta(M_k + M_k^\top) & \cos \theta(M_k - M_k^\top) \\ \cos \theta(M_k^\top - M_k) & \sin \theta(M_k + M_k^\top) \end{bmatrix} \prec 0, \quad (18)$$

282 $M_k = A_{cl,k}P$, $\zeta = \cos \theta$, places every closed-loop matrix of the convex hull
 283 in $\{\Re s < -\alpha\} \cap \{\zeta(s) > \zeta\}$ [9]. Bisection gives the boundary values; a 1 %
 284 safety factor is applied and the recovered P is re-checked for strictness.

285 *Discrete implementation.* All controllers run at T_s with a one-sample com-
 286 putational delay, total duty saturation and forward-Euler integral and back-
 287 calculation updates. Proposition 2(i) says that the derivative-filter state is an
 288 observer of the capacitor voltage, $x_F = cr - v_C$, and it should be discretized as
 289 one. The usual zero-order-hold discretization, $x_F[k+1] = \phi x_F[k] + (1-\phi)w[k]$
 290 with $w = cr - v_o$ and $\phi = e^{-NT_s}$, uses at sample k an estimate built from
 291 $v_o[k-1]$ only; during a 45 V slew v_o moves by about 2.7 V per sample and
 292 $K_d N \approx 0.7$ turns this lag into a duty error of order one. We therefore use
 293 the triangle-hold (first-order hold) update, exact when v_o is piecewise linear
 294 between samples,

$$x_F[k] = \phi x_F[k-1] + (1-\phi)w[k-1] + \gamma(w[k] - w[k-1]), \quad \gamma = 1 - \frac{1-\phi}{NT_s}, \quad (19)$$

295 evaluated with the current sample before the command $\tilde{d}[k] = d_{ff} + K_p(br[k] -$
 296 $v_o[k]) + x_I[k] + K_d N(w[k] - x_F[k])$ is formed, and initialized with $x_F[0] =$
 297 $cr[0] - v_C(0)$. Section 6.2 shows that this single choice decides whether the
 298 sampled PIDF remains equivalent to the sampled LQI.

299 4.5. Numerical instantiation

300 At the 24 V point, (5) evaluates to

$$\begin{aligned} A &= \begin{bmatrix} -1850.8259 & -9950.2488 \\ 9950.2488 & -995.0249 \end{bmatrix} \text{s}^{-1}, \\ B_d &= \begin{bmatrix} 483480 \\ 0 \end{bmatrix}, \quad C_y = [0.0497512 \quad 0.9950249]. \end{aligned} \quad (20)$$

301 For ξ_0 , $Q = \text{diag}(0.00444444, 0.00047259, 41978.4686)$, $R_d = 4.34027778$
 302 and (10) gives $K = [0.03564341 \quad 0.01212576 \quad -98.34550916]$ with a rel-
 303 ative Riccati residual of 2.18×10^{-14} and closed-loop poles -4243.01 and
 304 $-7917.86 \pm j6986.57 \text{ rad/s}$. The realization (14) gives

$$\begin{aligned} \theta_0 &= [K_p, K_i, K_d, N, b, c, K_b] \\ &= [0.0156901028, 98.3455092, 3.5037119 \times 10^{-6}, 2 \times 10^5, 1, 0, 9424.778], \end{aligned} \quad (21)$$

305 with a zero coefficient residual and a maximum relative frequency-domain
 306 error of 8.67×10^{-16} over $[10^{-1}, 10^8] \text{ rad/s}$. The minimum return difference is
 307 1.000000 and the phase margin of the full-state loop is 87.04° . The operating-
 308 polytope certificate at ξ_0 is $\alpha_{cert} = 4157.9286 \text{ rad/s}$ and $\zeta_{cert} = 0.7330911$
 309 ($\lambda_{\min}(P) > 0$ and strictly negative LMI eigenvalues were re-verified on the
 310 recovered matrices).

311 5. Constrained co-design problem and algorithms

312 5.1. Workflow

313 Fig. 3 summarizes the protocol. Offline, a candidate ξ is mapped by (10)
 314 and (14) to a 2-DOF PIDF; the averaged evaluator runs six scenarios and
 315 the switched screen runs one 34 ms transition sequence; Deb ordering ranks
 316 the record. After 240 global and 60 local calls the best record of each run is
 317 frozen, and only then replayed on the full switched mission, which is never
 318 visible to the search.

319 5.2. Scenarios and objective

320 The averaged evaluator integrates (3) with the hybrid guard $i_L \geq 0$ (RK4,
 321 $2 \mu\text{s}$) under the sampled control layer of Section 4. Six scenarios are declared
 322 (Table 4); each starts from the analytical equilibrium preceding its first event.
 323 For scenario s the seven criteria are the loss intensity E_{loss}/E_u , IAE, ISE,

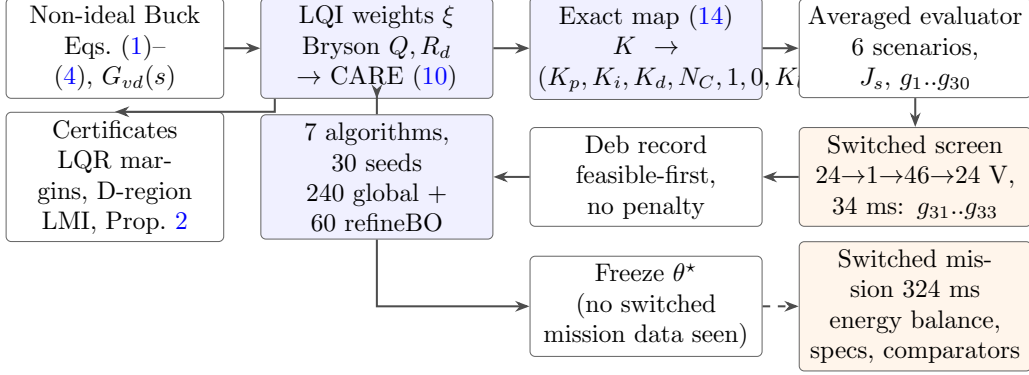


Figure 3: Formula-first CMO–LQI–PIDF workflow. Solid arrows: evaluation inside the search; dashed arrow: acceptance replay after the freeze. The switched screen is the only switched information used during the search.

Table 4: Averaged scenarios (the source is 48 V in all of them).

ID	T (ms)	reference	load pulse (+0.5 A)
S0	90	24→1→46 V at 30, 60 ms	15–25 ms
S1a	40	24→1 V at 10 ms	–
S1b	40	1→46 V at 10 ms	–
S1c	40	46→24 V at 10 ms	–
S2a	40	24 V	10–20 ms
S2b	40	46 V	10–20 ms

324 global ITAE, event-reset ITAE, RMSE and MAE, each divided by its value
325 for the normalization record ξ_0 . The scalar objective is the mean of the 6×7
326 ratios,

$$J_s(\xi) = \frac{1}{42} \sum_{s=1}^6 \mathbf{1}^\top r_s(\xi), \quad J_s(\xi_0) = 1. \quad (22)$$

327 MAE is IAE divided by the horizon and RMSE is the square root of ISE
328 divided by it, so two of the seven entries are redundant; the induced weights
329 (2/7 on the IAE family, 2/7 on the ISE family, 1/7 on each other criterion)
330 are declared rather than left implicit. The objective ranks feasible candidates
331 only.

332 *5.3. Thirty-three inequalities in engineering units*

333 For each scenario five inequalities $g_{s,j} \leq 0$ are evaluated separately:

$$\begin{aligned}
g_{s,1} &= \frac{\max_t \sigma_k (v_o - r_k)}{1.5 \text{ V}} - 1 && \text{overshoot beyond each new level,} \\
g_{s,2} &= \frac{\max_{t \in \mathcal{T}_p} |r - v_o|}{0.5 \text{ V}} - 1 && \text{load-pulse deviation,} \\
g_{s,3} &= \frac{\max_t i_L + \Delta I_{L,pp}^{\max}/2}{23 \text{ A}} - 1 && \text{device current,} \\
g_{s,4} &= \frac{\max_k |\bar{e}_k|}{25 \text{ mV}} - 1 && \text{steady error,} \\
g_{s,5} &= \frac{\rho_{\ell,s} - 1}{0.005} - 1, \quad \rho_{\ell,s} = \frac{E_{\text{loss}}/E_u}{(E_{\text{loss}}/E_u)_{\xi_0}} && \text{loss intensity,}
\end{aligned} \tag{23}$$

334 where σ_k is the sign of the k -th reference step, $\mathcal{T}_p$ runs from the onset of
335 the load pulse to the next event, $\bar{e}_k$ is the mean error over the last 2 ms of
336 each constant segment and 23 A is the IRF540N continuous drain rating at
337 100 °C. The averaged peak current is augmented by the worst half ripple
338 $\Delta I_{L,pp}^{\max}/2 = 1.17 \text{ A}$ (Table 3). A family that does not apply to a scenario (no
339 step, no load pulse) is set to -1 . The loss-intensity gate keeps the design
340 loss-aware: no scenario may convert energy less efficiently, per useful joule,
341 than the Bryson record ξ_0 by more than 0.5 %. These 30 inequalities are
342 completed by three measured ones,

$$g_{31} = \frac{\max_k \text{pp}(d)}{0.20} - 1, \quad g_{32} = \frac{\max_k |\bar{e}_{sw,k}|}{25 \text{ mV}} - 1, \quad g_{33} = \frac{\max_t i_{L,sw}}{23 \text{ A}} - 1, \tag{24}$$

343 obtained from one run of the switched replica over the transition sequence
344 $24 \rightarrow 1 \rightarrow 46 \rightarrow 24 \text{ V}$ (steps at 2, 14 and 24 ms, 34 ms in total, 1 μs RK4 with
345 exact PWM edges). The duty peak-to-peak and the mean error are measured
346 over the last 4 ms of each level. A candidate is feasible when $\sum_j \max(g_j, 0) \leq$
347 10^{-4} ; the screen costs about 12 ms per candidate against 14 ms for the six
348 averaged scenarios.

349 *Why the screen contains large transitions.* The protocol was frozen after two
350 pilot campaigns that are reported, not hidden. Pilot 1 used the analytical rip-
351 ple bound of the earlier version of this study, $\rho_{\text{rip}} = \max_V |C_{fb}(z=-1)| \Delta V_{o,pp}(V)/2 \leq$
352 0.20, where C_{fb} is the discrete feedback path evaluated at the controller
353 Nyquist frequency, which coincides with the carrier frequency because $T_s =$

354 $1/(2f_s)$. Every branch winner it produced satisfied the bound ($\rho_{rip} \approx 0.07$ –
 355 0.19) and yet, on the switched model, several sustained a limit cycle with
 356 duty swings of 0.7 – 0.9 at 24 V and offsets of 0.3 – 1.8 V. Pilot 2 replaced the
 357 bound by steady 10 ms switched windows started *from equilibrium* at 1 , 24
 358 and 46 V. Its best design (178 of 180 runs feasible) passed that screen, but on
 359 the reference mission it entered, after the large $24 \rightarrow 1$ V transition, a different
 360 limit cycle in which one sample in four clips at $d_{\min}$, leaving a -42 mV offset
 361 at 1 V. The loop thus holds *coexisting* switching limit cycles, and the one
 362 reached depends on the transient that precedes it; this is why the final screen
 363 is a transition sequence. ρ_{rip} is kept as a reported diagnostic only. Both pi-
 364 lots used the zero-order-hold filter; the triangle-hold update (19), adopted
 365 after the equivalence audit of Section 6.2, removes most of the limit-cycling
 366 designs by itself.

367 *Useful-energy band.* The earlier protocol also required $|E_u/E_{u,\xi_0} - 1| \leq 0.5\%$
 368 in every scenario. We report this quantity but no longer constrain it. An-
 369 chored to ξ_0 , it penalizes the removal of the 9.8 V undershoot of ξ_0 on the
 370 $46 \rightarrow 24$ V step, because a controller that holds 24 V delivers more energy than
 371 one that undershoots; anchored to the ideally regulated task, it penalizes
 372 every controller by 1 – 5% on step-down scenarios, where the asynchronous
 373 converter can only discharge through its load. With absolute overshoot, load-
 374 deviation and steady-error gates the delivered task is already pinned, and the
 375 loss intensity is normalized per useful joule, so under-delivery cannot improve
 376 it.

377 Feasibility is stated in absolute units, not as non-inferiority against ξ_0 .
 378 With ratio gates, 24 of 36 constraints are active at ξ_0 and a gentler normaliza-
 379 tion record leaves no candidate that dominates it; with absolute gates every
 380 controller family of Section 5.5 is judged against the same specification, and
 381 ξ_0 itself is infeasible because it overshoots by 6.2 V on the $1 \rightarrow 46$ V step.

382 5.4. Algorithms and budgets

383 Constraint handling uses Deb’s rules [11]: feasible beats infeasible, fea-
 384 sible records are ranked by J_s , infeasible ones by aggregate violation; no
 385 penalty weight is tuned. All methods work in the normalized box $u \in [0, 1]^n$,
 386 receive the matched seed $2026091200 + k$, $k = 1, \dots, 30$, and the same initial
 387 population of 12: ξ_0 first, six further individuals within a normalized radius
 388 0.05 of it, five uniform. One counted call is one complete evaluation (six

Table 5: Algorithms 1–7. Population 12, 240 global calls, then 60 refineBO calls initialized from the same run’s history (300 calls per run).

Alg.	Exploration / update	Exploitation / survival	Fixed parameters
1 PSO [21]	velocity with independent cognitive and social draws	personal and global Deb incumbents	$w = 0.9 - 0.5\tau$, $c_1 = c_2 = 1.7$, $ v \leq 0.2$
2 GA	binary Deb tournaments, SBX crossover	polynomial mutation, elitist ($\mu + \lambda$)	$\eta_c = 15$, $\eta_m = 20$, $p_m = 1/n$
3 ABC [20]	one-coordinate differential move (employed, onlooker)	greedy Deb replacement, scout	limit $\max(5, \text{round}(0.5 n_p n))$
4 pAEABC	ABC move scaled by diversity $s_k = \text{clip}(0.5 + 0.15/D_k, 0.5, 2)$, extra coordinate kick w.p. $0.25(1 - \tau)$	as ABC	inspired by Alsamia et al. [2]
5 pIGWO	three-leader GWO with $a = 2(1 - \tau^2)$, Gaussian kick w.p. $0.15(1 - \tau) + 0.02$	elitist pooling	inspired by Hou et al. [16]
6 pIGWO–DLH	linear $a = 2 - 2k/K$ GWO trial and an independently evaluated radial local-hunt trial	better trial replaces the wolf (non-elitist)	$R_i = \ x_i - x_\alpha\ /2$; inspired by Nadimi-Shahraki et al. [25]
7 refineBO	shared-kernel Matérn-5/2 GPs on J_s and $g_1, \dots, g_{33}$; pool of 1000 around the 10 Deb-best records, radius $0.04 \rightarrow 0.003$	$\alpha = P_f(\text{EI} + 0.02\sigma_J) + 10^{-4}P_f$, $P_f = \prod_j \Phi(-\mu_j/\sigma_j)$; restoration mode if no feasible incumbent	60 calls

averaged scenarios, the switched screen, 33 inequalities). Table 5 lists the operators.

The prefix “p” marks project variants that are inspired by, but not claimed to be operator-identical with, the cited algorithms. τ is the fraction of the budget consumed. The product P_f of marginal feasibility probabilities is an independence approximation used for ordering only; every accepted point is checked by the exact evaluator. Wall time is logged but never interpreted as an energy or performance quantity.

5.5. Controller families and fixed-rule comparators

To separate the value of the parameterization from that of the optimizer, the identical protocol (evaluator, 33 inequalities, 7 algorithms, 30 seeds, 300 calls) is run on six families:

- **CMO–LQI–PIDF**: ξ of (11) $\rightarrow$ (10) $\rightarrow$ (14); 4 variables, reads v_o only;
- **CMO–LQI–SF**: the same ξ and the same continuous gain, applied as full-state LQI with measured i_L and i_C (the twin of Proposition 2);
- **PIDF-direct**: the same 2-DOF structure ($b = 1$, $c = 0$, $N = N_C$) with $(\log K_p, \log K_i, \log K_d, \log K_b)$ searched directly over the gain ranges spanned by the LQI box;

- 408 • **PIDF7-direct**: $K_p, K_i, K_d, N, b, c, K_b$ all free (7 variables);
- 409 • **PID-direct**: backward-difference derivative on $-v_o$, $b = 1$, (K_p, K_i, K_d, K_b)
410 free;
- 411 • **PI-direct**: (K_p, K_i, K_b) free.

412 *Fixed-rule comparators.* Six classical designs share the control layer and are
413 synthesized at 24 V, $T_s = 10 \mu\text{s}$, with the delay $1.5T_s$ (computation plus
414 ZOH) in every frequency-domain rule. *PI*: crossover $0.03f_s$, phase margin
415 60° , $K_b = K_i/K_p$. *PID*: zeros cancelling the *LC* pair of $\Delta(s)$, crossover set for
416 60° of phase margin including the ESR zero, backward-difference derivative
417 on $-v_o$, $b = 1$, $c = 0$, $K_b = 1/\sqrt{T_i T_d}$. *PIDF*: the Type-III rule, i.e. the
418 PID zeros plus a filter pole on the ESR zero $N = N_C$, 60° of phase margin.
419 *LQR*: discrete Riccati solution on the ZOH model with the Bryson limits of
420 ξ_0 , measured i_L and $v_C = v_o - r_C i_C$, no integral action. *LQG*: the LQR gain
421 with a steady-state Kalman predictor on v_o only (process noise 0.01 in duty,
422 measurement noise 0.05 V). *LQI*: discrete Riccati solution of the integral-
423 augmented ZOH model with the same weights, measured state, $K_b = \omega_B$.
424 None of them is tuned by the proposed search; they represent what the
425 standard rules deliver on this plant.

426 5.6. Statistics and validation

427 Methods are compared as matched blocks over the 30 seeds with the
428 Friedman test and Holm-adjusted paired Wilcoxon signed-rank tests [12];
429 a run's score is J_s if feasible and 10^3 plus its violation otherwise, which is
430 consistent with Deb ordering. Families are compared pairwise on the same
431 (method, seed) cells. The retained controller of each family is the best feasible
432 record over its 180 runs; it is frozen and replayed on the 324 ms reference
433 mission of the Simulink model (24→1→46 V at 108 and 216 ms, +0.5 A load
434 pulse at 54–90 ms, initial state at the 24 V equilibrium), with the energy
435 balance of (8).

436 6. Results and discussion

437 6.1. Formula-first verification

438 Every analytical quantity of Sections 3–4 was recomputed twice: from the
439 closed forms and from independent numerical routes (state-space to transfer-
440 function conversion, a general CARE solver, a frequency sweep of (12) against

Table 6: Structural-equivalence audit (Proposition 2): largest $|v_o^{LQI} - v_o^{PIDF}|$ over the 324 ms mission.

Case	predicted	measured
nominal, $i_p = 0$	identical	1.6×10^{-13} V
$L - 20\%$, $r_L \times 2$, $R_{on} \times 2$	identical	2.8×10^{-14} V
0.5 A load pulse	$\neq$, Eq. (17)	0.411 V
$R + 20\%$	$\neq$, Eq. (17)	0.609 V
$C + 20\%$	$\neq$, $N \neq N_C$	2.823 V
sampled, ZOH filter, 1→46 V	close	14.54 V
sampled, FOH filter (19), 1→46 V	close	0.35 V

(13), and an SDP solver for (18)). The results of Section 4.5 agree with the values archived by the earlier version of this study, obtained with a different code base: the gain, the poles and θ_0 to all printed digits, and the LMI bounds α_{cert} , ζ_{cert} to within 2×10^{-7} relative, the bisection tolerance. The companion MATLAB script `cmo_formula_first.m` repeats the chain with `icare` and `freqresp`.

6.2. The equivalence, tested where it should hold and where it should fail

Table 6 and Fig. 4 test Proposition 2 on the 324 ms reference mission with continuous-time controllers on the nonlinear averaged plant (RK4, 0.05 μ s). With nominal output network and no load pulse the LQI and its 2-DOF PIDF produce identical trajectories to 1.6×10^{-13} V, although 1.1 % of the mission is spent in duty saturation, the averaged DCM guard $i_L \geq 0$ is active and the anti-windup path is exercised. Changing L by -20% and doubling r_L and R_{on} leaves the identity intact (2.8×10^{-14} V, with 34.1 % of saturation), exactly as the proof predicts, because these parameters never enter the output network. The three predicted failure modes appear with the predicted signs: a 0.5 A load pulse (0.411 V), a 20 % load mismatch (0.609 V) and a 20 % capacitance mismatch that detunes N from N_C (2.823 V).

Sampling is the practical case. With the zero-order-hold filter the sampled PIDF departs from the sampled full-state LQI by 14.54 V on the 1→46 V step; with the triangle-hold update (19) the departure falls to 0.35 V, while both sampled laws sit 2.54 V from the continuous trajectory because of the common one-sample delay. The realization is therefore exact in continuous time and remains a close realization under sampling *provided* the filter is discretized as the capacitor-voltage observer it is.

Table 7: CMO–LQI–PIDF family: 30 matched seeds per algorithm (Algorithms 1–6, each followed by refineBO). J_s statistics over feasible runs; “global counts runs already feasible after the 240 global calls.

Algorithm	feasible	global	best J_s	median	IQR	mean rank
PSO	18/30	18/30	0.8152	0.8283	0.0208	3.77
GA	13/30	13/30	0.8151	0.8330	0.0190	4.23
ABC	17/30	8/30	0.8152	0.8163	0.0162	3.77
pAEABC	25/30	18/30	0.8151	0.8161	0.0177	2.77
pIGWO	28/30	27/30	0.8153	0.8308	0.0205	3.10
pIGWO-DLH	23/30	18/30	0.8152	0.8291	0.0256	3.37

6.3. Seven algorithms on the LQI-weight space

Table 7 and Fig. 5(a) summarize the 180 runs of the CMO–LQI–PIDF family. 124 runs end feasible. All six branches converge to the same optimum: their best records lie between 0.8151 and 0.8153, a spread of 2×10^{-4} , and decode to practically the same controller. The LQI-weight space therefore has a single, well-defined optimum under the 33 inequalities, and the retained design is not an artefact of one operator or one seed. The retained record improves the normalized objective by 18.5 % with respect to the Bryson record ξ_0 , while ξ_0 itself violates the overshoot specification.

What separates the operators is reliability, not the optimum. The Friedman test over matched seeds rejects equality ($Q = 11.96$, $p = 0.035$); after Holm correction 1 pairwise difference remains (GA–pIGWO). pIGWO reaches a feasible design in 28 of 30 seeds and pAEABC in 25, against 13 for GA. The refinement stage matters most where the global stage is weakest: it raises the number of feasible runs from 8 to 17 for ABC and from 18 to 25 for pAEABC (Table 7), whereas it adds one run for pIGWO and none for PSO and GA. A practitioner who can afford only one run should therefore prefer the grey-wolf or adaptive-bee variant followed by refineBO; one who can afford a few seeds can use any of the six.

Retained controller. The retained weights are $\Delta_i = 10.842$ A, $\Delta_v = 3.301$ V, $\omega_B = 6505$ rad/s, giving $K = [0.090893, 0.165271, -945.806]$ and closed-loop poles -6493 and $-20149 \pm j17169$ rad/s. Its exact realization is

$$\theta^* = [K_p, K_i, K_d, N, b, c, K_b] = [0.17436, 945.81, 8.2629 \times 10^{-6}, 2 \times 10^5, 1, 0, 92951]. \quad (25)$$

488 Being an LQI law, it inherits the full-state LQR margins (minimum return
489 difference 1, phase margin 70.25°), and the common- P certificate over the
490 operating polytope gives $\alpha_{cert} = 6414.8 \text{ rad/s}$ and $\zeta_{cert} = 0.7496$, i.e. 1.54
491 times the guaranteed decay rate of ξ_0 at slightly better guaranteed damping.
492 None of these certificates is available for a gain vector found by direct search.

493 *6.4. Does the LQI parameterization help? Six families, one protocol*

494 Table 8 and Fig. 5(b) compare six families run under the identical pro-
495 tocol, so that the only difference between two rows is the space in which the
496 seven algorithms search.

497 *Same structure, different coordinates.* PIDF-direct searches the same 2-DOF
498 PIDF ($b = 1$, $c = 0$, $N = N_C$) directly in $(\log K_p, \log K_i, \log K_d, \log K_b)$ over
499 the gain ranges spanned by the LQI box. It reaches the same optimum
500 (best $J_s = 0.8154$ against 0.8151) with a similar number of feasible runs
501 (129 against 124 of 180), and the paired comparison on the same (algorithm,
502 seed) cells is not significant ($p = 0.74$). The LQI coordinates therefore do
503 not make the search easier on this problem. What they change is what
504 every candidate is. Inverting the map (14) and applying Kalman’s return-
505 difference condition shows that the best direct-search design is itself LQ-
506 optimal, $K = [0.0906, 0.1650, -943.2]$ with $\min_\omega |1 + L_K(j\omega)| = 1$, i.e. the
507 gain search converges onto the LQ manifold; yet only 75 of its 129 feasible
508 designs satisfy the condition, so 42 % of the controllers an equally lucky gain
509 search may return carry no LQ optimality, no guaranteed margins and no
510 reason to be certifiable. The LQI parameterization confines the search to
511 that manifold from the first evaluation, at no cost in optimality.

512 *Richer and poorer structures.* PID-direct (backward-difference derivative)
513 reaches best $J_s = 0.8249$ in 125 feasible runs and is not significantly dif-
514 ferent from CMO–LQI–PIDF ($p = 0.41$). PIDF7-direct, with free filter pole
515 and setpoint weights, finds more feasible runs (163) but a worse optimum
516 (0.8513), with $N = 141939 \text{ rad/s}$, $b = 1.00$ and $c = 0.31$; on the switched mis-
517 sion this design holds the 1 V level with 0.0 mV of error and IAE 0.0348 V s,
518 i.e. the extra freedom buys switched robustness at the price of the averaged
519 objective. PI-direct never becomes feasible (minimum violation 1.04): a PI
520 cannot meet the overshoot, load-deviation and current specifications of the
521 six scenarios at once, and the paired comparison is decisive ($p = 2.7 \times 10^{-31}$).

Table 8: Controller families under the identical protocol (7 algorithms, 30 seeds, 300 calls, 33 inequalities). Paired comparison with CMO–LQI–PIDF on the same (algorithm, seed) cells: Deb-consistent score, two-sided Wilcoxon signed-rank test.

Family	variables	feasible runs	best J_s	median J_s (feasible)	CMO better / tie / worse	p	min. violation
CMO-LQI-PIDF	4 (LQI weights, K_b)	124/180	0.8151	0.8287	–	–	0.000
CMO-LQI-SF	4 (same, full state)	154/180	0.9054	0.9749	132 / 0 / 48	3.5×10^{-2}	0.000
PIDF-direct	4 (K_p, K_i, K_d, K_b)	129/180	0.8154	0.8273	99 / 0 / 81	7.4×10^{-1}	0.000
PIDF7-direct	7 (+ N, b, c)	163/180	0.8513	1.1018	121 / 0 / 59	7.7×10^{-1}	0.000
PID-direct	4	125/180	0.8249	0.8278	99 / 0 / 81	4.1×10^{-1}	0.000
PI-direct	3	0/180	–	–	180 / 0 / 0	2.7×10^{-31}	1.041

522 *The full-state twin.* CMO–LQI–SF uses the same four variables but applies
523 the gain as a measured-state LQI. It is feasible more often (154 runs) but
524 at a worse averaged optimum (0.9054) than its output-feedback realization
525 ($p = 0.035$ in favour of the PIDF on the averaged objective). On the switched
526 mission the ordering reverses (Section 6.5): the twin is the more robust
527 controller at the 1 V level, which is the switched counterpart of the output-
528 current term of (17).

529 *What the protocol, rather than the parameterization, buys.* Every family tuned
530 under the constrained multi-fidelity protocol ends close to full compliance on
531 the reference mission (Table 9: u between 0.74 and 1.02; the lowest value
532 belongs to a slow PI whose IAE, 0.0943 V s, is 2.6 times that of the others),
533 whereas the six fixed-rule designs lie between 6.28 and 30.17. The decisive
534 ingredients are the engineering-unit specification and the switched transition
535 screen; the choice of search coordinates decides the guarantees that come
536 with the result.

537 6.5. Switched validation on the reference mission

538 Table 9 and Fig. 6 report the frozen controllers on the 324 ms mission of
539 the reference model. The energy balance closes to better than 4.5×10^{-6} % of
540 the input energy for every row, so all efficiency and loss figures are resolved
541 by the integrator. As a fidelity check against the Simulink audit of the earlier
542 version of this study, the replica gives for the fixed PI 26.63 V of overshoot,
543 40.78 A and IAE 0.0480 V s (Simulink: 26.75 V, 40.76 A, 0.0464 V s) and for
544 the fixed LQI 9.41 V, 32.61 A and 0.0348 V s (Simulink: 9.56 V, 31.27 A,
545 0.0346 V s).

546 *Answer to the comparison question.* Against the six classical designs the
547 result is unambiguous on the specification, and nuanced on individual indices.

Table 9: Switched replay on the 324 ms reference mission (24→1→46 V, 0.5 A load pulse). Specification ratios: overshoot/1.5 V, load deviation/0.5 V, peak i_L /23 A, worst steady error/25 mV, steady duty peak-to-peak/0.20; a ratio ≤ 1 meets the specification. u is the largest ratio. Energy residual below 4.5×10^{-6} % for every row.

Controller	IAE (Vs)	overshoot (V)	$i_{L,pk}$ (A)	$ e _{\max}$ (mV)	load dev. (V)	η (%)	E_{loss}/E_u	sat. frac.	u	specs met	sensors
CMO-LQI-PIDF (retained, v_o only)	0.0364	0.02	22.44	25.6	0.137	97.978	0.01801	0.089	1.02	4/5	v_o
CMO-LQI-SF (twin, measured i_L)	0.0347	0.02	22.87	0.0	0.214	97.978	0.01800	0.008	0.99	5/5	v_o, i_L, i_C
PIDF, direct gain search	0.0363	0.02	22.42	25.0	0.137	97.978	0.01801	0.089	1.00	5/5	v_o
PIDF7, direct search	0.0348	0.02	22.11	0.0	0.164	97.978	0.01800	0.011	0.96	5/5	v_o
PID, direct search	0.0368	0.02	21.83	23.7	0.138	97.980	0.01798	0.087	0.95	5/5	v_o
PI, direct search	0.0943	0.23	6.04	6.8	0.369	97.987	0.01786	0.009	0.74	5/5	v_o
Bryson LQI-PIDF ξ_0	0.0340	8.05	32.50	0.0	0.193	97.936	0.01845	0.012	5.37	3/5	v_o
PI (loop shaping)	0.0480	26.63	40.78	0.1	0.474	97.813	0.01972	0.012	17.76	3/5	v_o
PID (pole-zero)	0.0467	28.64	40.77	11.3	0.128	97.802	0.01984	0.093	19.09	3/5	v_o
PIDF (Type III)	0.0453	28.65	40.78	0.0	0.157	97.801	0.01986	0.013	19.10	3/5	v_o
LQR	0.1339	0.97	24.36	754.3	0.805	97.969	0.01811	0.001	30.17	2/5	v_o, i_L, i_C
LQG	0.0716	0.28	24.39	289.5	0.466	97.960	0.01818	0.000	11.58	3/5	v_o
LQI (discrete, full state)	0.0348	9.41	32.61	0.0	0.552	97.931	0.01849	0.013	6.28	2/5	v_o, i_L, i_C

- *PI, PID and PIDF* designed by standard frequency-domain rules hold every level but overshoot the 1→46 V step by 26.63–28.65 V and draw about 40.78 A, i.e. 1.8 times the 23 A device rating; their specification ratios are 17.76–19.10.
- *LQR and LQG*, having no integral action, leave offsets of 754.3 and 289.5 mV at the worst level.
- *LQI*, synthesized from the same Bryson limits and reading the full state, tracks well (IAE 0.0348 Vs) but overshoots by 9.41 V and peaks at 32.61 A.
- The retained *CMO-LQI-PIDF*, reading v_o only, removes the overshoot, stays at 22.44 A, keeps the load deviation at 0.137 V and converts at 97.978 %, against 97.80–97.97 % for the six fixed-rule designs. Its IAE, 0.0364 Vs, is within 5 % of the LQI and 7 % of ξ_0 , whose lower IAE is bought with 8–9 V of overshoot. It misses one of the five mission specifications by 0.6 mV: 25.6 mV of steady error at 1 V for a 25 mV limit ($u = 1.02$).
- Its full-state twin *CMO-LQI-SF*, with the same weights and a measured inductor current, meets all five specifications ($u = 0.99$) with IAE 0.0347 Vs.

The procedure is therefore superior to the six classical design rules in the sense that matters to an industrial user: every controller it returns, in any of the tuned families of Table 8, removes the overshoot, stays within the

device rating and holds the levels, with specification ratios between 0.74 and 1.02 against 6.28–30.17 for PI, PID, PIDF, LQR, LQG and LQI. It does not dominate every index: the LQI and ξ_0 track slightly better in IAE, at the cost of 8–9 V of overshoot and 32–33 A of peak current. The retained CMO–LQI–PIDF and its direct-search counterpart are practically the same controller (IAE 0.0364 and 0.0363 V s), and their 1 V steady error sits on the 25 mV line (25.6 and 25.0 mV); this level is the binding one, which the switched screen resolves only to within a millivolt because it is a shorter transient than the mission.

The value of the current sensor. Among the LQI-structured controllers, the full-state twin meets every specification and the output-feedback realization misses one by 0.6 mV. By Proposition 2 the two differ only by the term $-K_{ix}(i_o - v_o/R)$ of (17), an instantaneous feedback on the unmodelled output current. At 1 V in discontinuous conduction the output-current estimate v_o/R is least accurate relative to the signal, and the PIDF must remove the resulting bias through its integrator in the presence of a switching limit cycle; the twin removes it directly. The gap between the two rows of Table 9 is thus the measured dividend of one current sensor, predicted in sign and location by the theory. The efficiencies of all tuned designs agree within 0.01 percentage point (97.978–97.987 %), so the conversion loss does not discriminate between them; it does separate them from the fixed-rule designs, whose overshoots and current peaks cost 0.01–0.18 percentage point.

6.6. Limitations

The evidence is simulated. The switched replica reproduces the controller-independent figures of the reference Simulink model and its fixed comparators, but the final numbers of Table 9 should be regenerated on the `.slx` with the provided S-function before being quoted as Simulink results. Losses are the modelled conduction and snubber losses of the declared boundary; dynamic switching loss, reverse recovery, magnetic and thermal effects are outside it, and the efficiency differences between rows (a few hundredths of a percentage point) are far below the accuracy of a laboratory power analyser. Component tolerances were not supplied and are not claimed: the common- P certificate covers the nominal operating polytope only. The mission and the specification are those of a fixed 48 V source; line regulation is outside the study. Finally, the specification set and the filter discretization were revised

during pilot campaigns that are reported in Section 5 and Appendix Appendix B; the final campaign was run once, with the protocol frozen, and its switched validation was not used for selection.

7. Conclusions

A formula-first co-design was carried out for a non-ideal asynchronous Buck converter supplied at 48 V and commanded over 1 V to 46 V. Its search space is the set of Bryson excursions of an LQI cost, so every evaluated point is a Riccati-optimal law with LQR margins and an operating-polytope certificate, and every point is implemented exactly as a 2-DOF PIDF that measures the output voltage only.

Three results carry the paper.

Theory. The realization is structural, not only nominal: the derivative filter of the PIDF, with its pole at the ESR zero, reconstructs the capacitor voltage along every trajectory of the converter, so the PIDF and the full-state LQI coincide under saturation, anti-windup and discontinuous conduction and for any inductor-branch or semiconductor parameters. They differ only by an explicit feedback on the unmodelled output current, by a mismatch of the output network, and by sampling. Under sampling the equivalence survives when the filter is discretized as the observer it is: a triangle-hold update keeps the sampled PIDF within 0.35 V of the sampled LQI on a 45 V step, where the usual zero-order-hold update departs by 14.54 V.

Optimization. Seven algorithms applied to the four LQI weights converge to one optimum; they differ in reliability, pIGWO and pAEABC reaching feasibility in 28 and 25 of 30 seeds against 13 for GA, and the Gaussian-process refinement matters most where the global stage is weakest. Searching the same 2-DOF PIDF directly in gain space reaches the same optimum, which turns out to be LQ-optimal; the LQI parameterization does not accelerate the search on this problem, but it confines it to the manifold on which the optimum lies and gives every candidate its margins and certificate, whereas 42 % of the feasible direct-search designs are not LQ-optimal for any weight.

Validation. Switched admissibility is not inherited from an averaged model: an analytical ripple bound and a steady switched screen both admitted designs that limit-cycle on the reference mission, because the loop holds coexisting switching limit cycles selected by the preceding transient. A 34 ms switched transition screen inside the search removes them. On the reference mission, the retained CMO-LQI-PIDF removes the overshoot of

the classical designs and stays within the 23 A device rating; its only miss is a steady error of 25.6 mV at 1 V against a 25 mV limit, which its full-state twin meets. PI, PID, PIDF, LQR, LQG and LQI designed by their standard rules exceed at least two specifications by factors of 6 to 30.

The contribution is therefore a design procedure an industrial engineer can apply with standard tools: choose admissible excursions, solve one Riccati equation, realize the gain as a PIDF with setpoint weight one and a derivative pole on the ESR zero, discretize the filter as a capacitor-voltage observer, and let a small, constrained, multi-fidelity search tune four physically meaningful numbers. Hardware validation, component-tolerance robustness and line regulation are the next evidence levels.

Appendix A. Reproducibility record

Reference model. buck_fo_lqi_fixed_reference_v6_1_46V_R2022a.slx, SHA-256 948fbae2679b57ed726a92eeebcae398aaf0ca6b9b3a74216759ca80b390c7fe.

All plant, mission and logging parameters of Table 2 were decoded from its model workspace; the snubber is resistive ($R_s = 100 \text{ k}\Omega$, $C_s = \infty$) on the MOSFET only, the diode snubber is disabled, and the MOSFET body diode has $R_d = 0.01 \Omega$, $V_{fd} = 0$. The saved model initializes L at 1.2 A and C at 12 V; all replays of this paper start at the 24 V equilibrium (2.4 A, 24 V).

Commands. From the repository root:

```
python scripts/cmo_formula_first.py    # Sections 3-4, Prop. 2 tests
python scripts/cmo_campaign.py         # 6 families x 6 methods x 30 seeds
python scripts/cmo_analyze.py          # statistics, freeze, switched replay
python scripts/cmo_figures.py; python scripts/cmo_numbers.py
```

In MATLAB (R2022a or later, Simscape Electrical, Control System Toolbox):

```
cmo_formula_first
T = run_cmo_validation(slx, 'results/cmo/controllers.csv');
```

The second call replaces only the controller S-function (`cmo_sfun`, a line-by-line port of `cmo/ctrl.py`) and never saves the model.

Seeds and budgets. Seeds 2026091201, ..., 2026091230; population 12; 240 global and 60 refineBO calls per run; 30 runs per method and family. Each checkpoint stores the method, seed, best decision vector, objective, the 33 inequalities, the global-stage record and the best-so-far curve.

675 **Appendix B. Protocol changes with respect to the earlier version**

676 **Data availability**

677 All code (Python replica, evaluator, algorithms, analysis, figures), the
678 MATLAB S-function and replay script for the reference Simulink model,
679 the per-run checkpoints and the generated tables are archived with the
680 manuscript repository; a public DOI will be assigned at acceptance.

681 **Declaration of competing interest**

682 The authors declare that they have no known competing financial inter-
683 ests or personal relationships that could have appeared to influence the work
684 reported in this paper.

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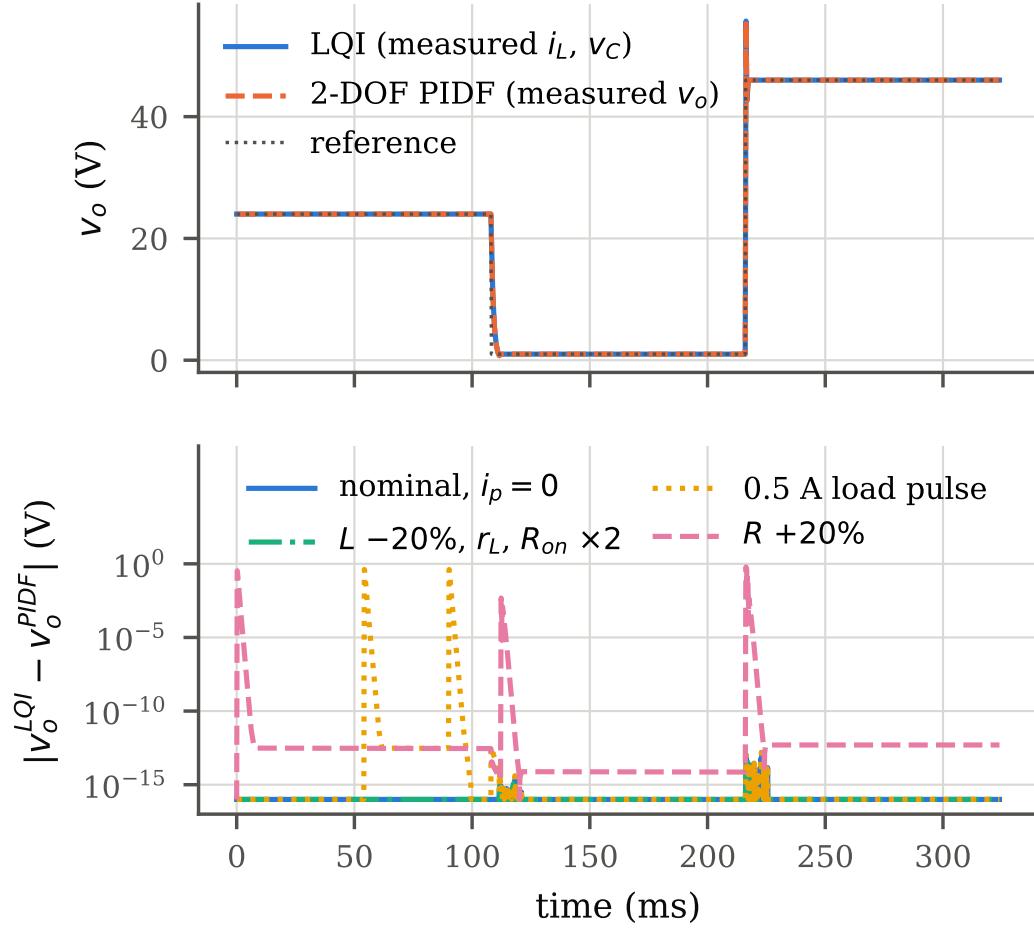


Figure 4: Top: LQI with measured (i_L, v_C) and its 2-DOF PIDF with measured v_o on the reference mission (continuous time, averaged nonlinear plant, saturation active). Bottom: absolute output difference; the nominal and inductor-branch-mismatch curves lie at the floating-point floor, the load-pulse and load-mismatch curves show the predicted departures.

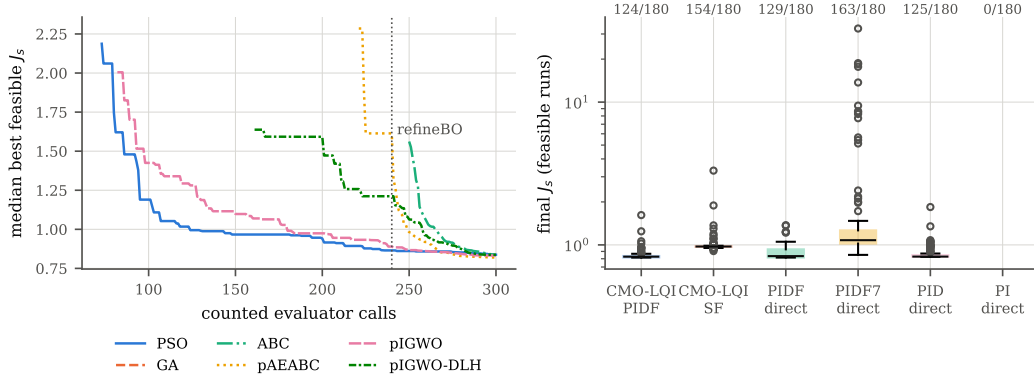


Figure 5: (a) Median best-feasible objective against counted calls for the CMO-LQI-PIDF family (dotted line: end of the global stage); a curve starts once at least half of the 30 seeds hold a feasible record, which GA never achieves (13/30). (b) Final objective of the feasible runs of every family under the identical protocol, with the number of feasible runs out of 180 above each box.

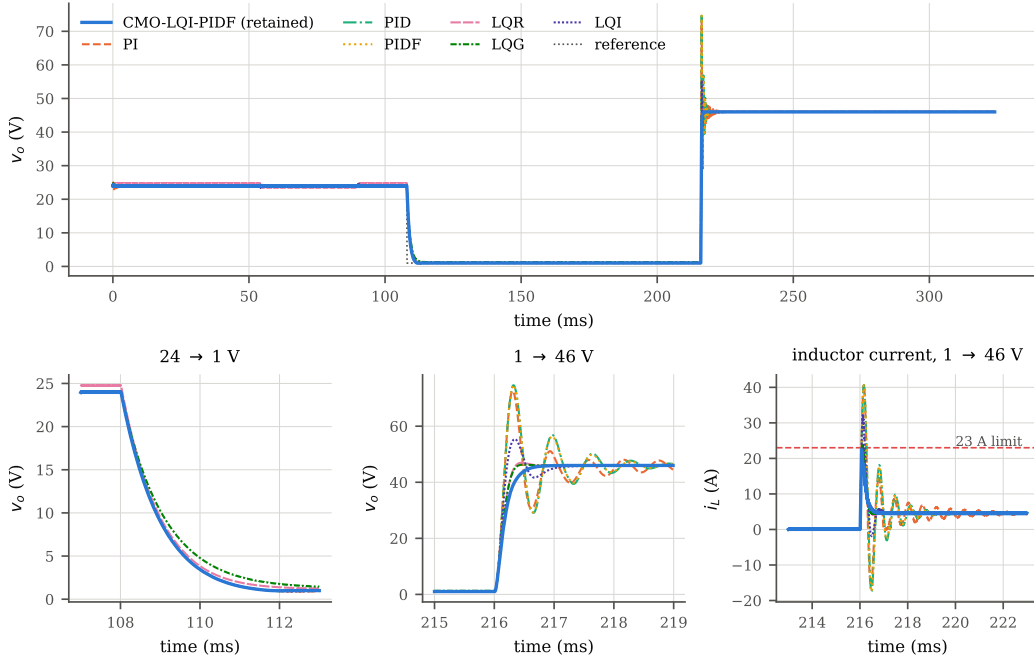


Figure 6: Switched replay on the reference mission: retained CMO-LQI-PIDF and the six fixed-rule comparators. Bottom: 24→1 V and 1→46 V transitions and the inductor current on the 1→46 V step against the 23 A device limit.

Table B.1: Declared changes and their reasons.

Change	Reason
Feasibility in engineering units instead of ratios to ξ_0	with ratio gates 24 of 36 constraints are active at ξ_0 , and a gentle record leaves no dominating candidate; absolute gates make all families comparable
Analytical ripple bound $\rightarrow$ switched transition screen (g_{31} – g_{33})	pilots 1 and 2: the bound and a steady screen both admitted limit-cycling designs (Section 5)
Device current limit 23 A (IRF540N at 100 °C)	the earlier retained design used 90 % of the 25 °C rating
refineBO budget stated once: 60 calls	the earlier text gave both 20 and 60
Structural equivalence (Prop. 2)	replaces the claim that saturation, DCM and PWM break the realization
Triangle-hold (FOH) discretization of the derivative filter, Eq. (19)	the ZOH filter makes the sampled PIDF deviate by up to 14.54 V from the sampled LQI on the 1 $\rightarrow$ 46 V step; FOH reduces it to 0.35 V (identified by the equivalence audit, before the final campaign)
Useful-energy band reported, not constrained	ill-posed on step-down transients of an asynchronous converter (Section 5); identified before the final campaign
Six added families and six fixed-rule comparators on one specification	separates parameterization from optimizer